

Oblig 1 - Sondre Fosså

1 Detuctive reasoning

Exercise 1

A Since G is false, one of E, F and B must be false. A is true, so B is true. If E were true, F would also be true, so E must be false. Since A is true and E is false, D must be false. Since A and B are true, C must also be false. F can be either true or false since E is false.

B B is true, rest is false B is true from the first premise which means one of D and E have to be false If either is false the other one is which means both are false

Exercise 2

A The assumptions are not enough to guarantee that C / P is valid or invalid. If $P = C = p$, then C / P is valid. If $P = p$ and $C = p \vee q$, then C / P is invalid.

B Under the given assumptions, P / Q is valid.

Since $P \models C$ and $C \models Q$, transitivity gives $P \models Q$, which makes P / Q valid.

C Under the given assumptions, Q / P is NOT valid.

If Q / P was valid, then $Q \models P$. Since $P \models C$, transitivity would give $Q \models C$, which contradicts $Q \not\models C$. Therefore Q / P is not valid.

Propositional logic

Exercise 3

Syntactic ambiguity means that a formula can be grouped in more than one way. $p \wedge q \vee r$ can mean $(p \wedge q) \vee r$ or $p \wedge (q \vee r)$. $p \rightarrow q \rightarrow r$ can mean $(p \rightarrow q) \rightarrow r$ or $p \rightarrow (q \rightarrow r)$.

Exercise 4

A $\neg p = p \rightarrow \perp$

B $p \wedge q = (p \rightarrow (q \rightarrow \perp)) \rightarrow \perp$

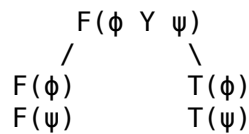
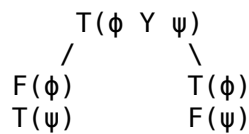
C $p \vee q = (p \rightarrow \perp) \rightarrow q$

D $p \leftrightarrow q = ((p \rightarrow q) \rightarrow \perp) \rightarrow (q \rightarrow p)$

Exercise 5

A $(\phi \vee \psi) \wedge \neg(\phi \wedge \psi)$

B



Exercise 6 - From natural language to the predicate language

Predicates:

- $\text{Boy}(x)$: x is a boy
- $\text{Girl}(x)$: x is a girl
- $\text{Loves}(x, y)$: x loves y

A Everybody loves somebody.

$$\forall x \exists y \text{ Loves}(x, y)$$

B Some boy doesn't love all girls.

$$\exists x (\text{Boy}(x) \wedge \exists y (\text{Girl}(y) \wedge \neg \text{Loves}(x, y)))$$

C Every boy who loves a girl is also loved by some girl.

$$\forall x ((\text{Boy}(x) \wedge \exists y (\text{Girl}(y) \wedge \text{Loves}(x, y))) \rightarrow \exists z (\text{Girl}(z) \wedge \text{Loves}(z, x)))$$

D Every girl who loves all boys does not love every girl.

$$\forall x ((\text{Girl}(x) \wedge \forall y (\text{Boy}(y) \rightarrow \text{Loves}(x, y))) \rightarrow \exists z (\text{Girl}(z) \wedge \neg \text{Loves}(x, z)))$$

E No girl who does not love a boy loves a girl who loves a boy.

$\forall x ((\text{Girl}(x) \wedge \exists y (\text{Boy}(y) \wedge \neg \text{Loves}(x, y))) \rightarrow \forall z ((\text{Girl}(z) \wedge \exists w (\text{Boy}(w) \wedge \text{Loves}(z, w))) \rightarrow \neg \text{Loves}(x, z)))$

F Some boy loves all girls who love only girls.

$\exists x (\text{Boy}(x) \wedge \forall y ((\text{Girl}(y) \wedge \forall z (\text{Loves}(y, z) \rightarrow \text{Girl}(z))) \rightarrow \text{Loves}(x, y)))$

Exercise 7

A There exists a person such that everyone they love, loves them back.

B Every girl is loved by someone

C There is no girl who loves noone

D Everyone who loves someone is loved by someone

E Someone loves everyone

F Every girl who loves at least one girl is loved by at least one boy

Exercise 8

A $D = \{a, b, c\}$ $R = \{(a \rightarrow b), (b \rightarrow c), (c \rightarrow b)\}$

A is true since everyone points to someone. B is false since no one points to a, and C and D are false since no one points to everyone or is pointed to by everyone.

B $D = \{a, b, c\}$ $R = \{(b \rightarrow a), (c \rightarrow b), (b \rightarrow c)\}$

B is true since everyone is pointed to by someone. A is false since a points to no one, and C and D are false since no one points to everyone or is pointed to by everyone.

C C cannot be true while B is false. If one x points to every y, then every y is pointed to by someone, which makes B true.

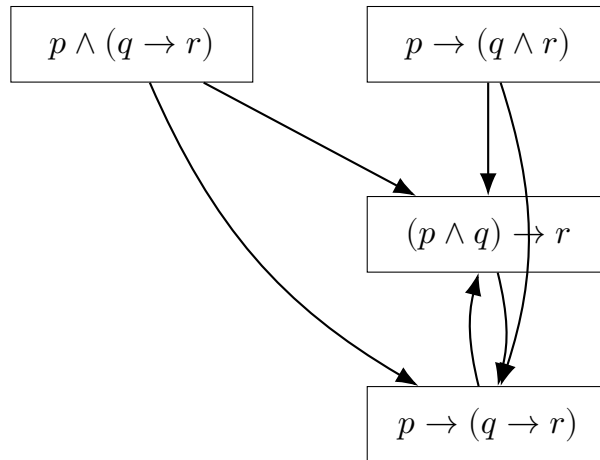
D D cannot be true while A is false. If every x points to one y, then every x points to someone, which makes A true.

Exercise 9

A $\phi \wedge \neg\phi / \psi$ is valid for any ϕ, ψ . Since the inference is only invalid if the premises are true but the conclusion is false, the premise can never be true so the inference is always valid.

B $\phi / \psi \vee \neg\psi$ is valid for any ϕ, ψ . The conclusion is always true so the inference is always valid.

Exercise 10



Exercise 11

A $\forall x.(P(x) \vee Q(x)) / \forall x.P(x) \vee \exists x.Q(x)$ is valid. If $\forall x(P(x))$ is true then the conclusion is already true. If $\forall x(P(x))$ is false then $\forall x(Q(x))$ must be true which means that the conclusion is also true.

B $\exists x.(P(x) \wedge Q(x)) / \exists x.P(x) \wedge \exists x.Q(x)$ is valid.

For a given x if both P(x) and Q(x) are true then the conclusion is true.

C $\exists x.P(x) \wedge \exists x.Q(x) / \exists x.(P(x) \wedge Q(x))$ is not valid.

$D = \{a, b\}$, $P = \{a\}$, $Q = \{b\}$. The premise is true, but no object is both P and Q.

D $\exists x . P(x) \wedge \forall x . Q(x) / \exists x . (P(x) \wedge Q(x))$ is valid.

Take an object where P is true. Since Q is true for every object, Q is also true for some objects

E $\forall x . \forall y . (R(x, y) \rightarrow \neg R(y, x)) / \forall x . \neg R(x, x)$

The premise says that the relation can only be one way therefore you cant have x to x aswell so the inference is valid.

F $\forall x . P(x) \rightarrow \forall x . Q(x) / \forall x . (P(x) \rightarrow Q(x))$ is not valid.

$D = \{a, b\}$, $P = \{a\}$, $Q = \{\}$. The premise is true since not every object is P, but the conclusion is false since $P(a)$ is true and $Q(a)$ is false.

5 Default reasoning

12

A If ψ already follows from Φ in normal logic, then it will still follow when CWA adds extra assumptions. So: Yes

B If ψ follows from Φ in CWA logic, then it is not guaranteed to follow in normal logic. $\Phi = \{q\}$ $\psi = \neg p$ In CWA we assume p to be false, since we dont know anything about it which makes the inference valid. But its invalid in normal logic.

13

A Yes.

If $\Phi \models \psi$, then ψ is true in every model of Φ . The most plausible models are just some of those models.

B Yes. If there were a model of Φ where ψ was false, we could make that model most plausible. Then the plausibility consequence would fail.

C Yes. A says it works for every ordering like in A so it works for some

D No, because you can make the models where ψ is true the most plausible, even though there are less plausible models of Φ where ψ is false.